



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The multimedia company suffered a Denial of Service (DoS) attack via an ICMP flood. A malicious actor exploited an unconfigured firewall to send a massive volume of ICMP ping requests, overwhelming the network and cutting off all internal services for approximately two hours. The incident management team responded by blocking ICMP traffic, shutting down non-critical services, and restoring critical network operations. New protective measures including updated firewall rules, IP verification, and network monitoring tools were then implemented.
Identify	Attack Type: ICMP Flood / Denial of Service (DoS) attack. Systems Affected: The company's entire internal network infrastructure and all network-dependent services (web design, graphic design, and social media marketing platforms). Root Cause: An unconfigured firewall failed to filter or rate-limit incoming ICMP packets, allowing the attacker to overwhelm the network and deny access to all internal users.
Protect	The firewall must be properly configured with rules to limit the rate of incoming ICMP packets and block traffic from suspicious or unverified

	sources. Source IP address verification should be enforced on the firewall to detect spoofed addresses. Staff should receive training on recognizing early signs of DoS attacks. Policies around firewall management and network access control must be updated to prevent misconfigured systems from being exploited in the future. An IDS/IPS system should be implemented to filter suspicious ICMP traffic.
Detect	Network monitoring software should be deployed to continuously analyze traffic patterns and flag abnormal spikes in ICMP activity. An Intrusion Detection/Prevention System (IDS/IPS) should be configured to identify and alert on suspicious ICMP floods and non-trusted IP sources attempting to pass through the firewall. A SIEM tool can be used to centralize alerts and enable faster detection and response by the security team.
Respond	Upon detecting a similar attack, the team should immediately block all malicious ICMP traffic at the firewall, isolate affected systems, and take non-critical services offline to protect critical operations. Incident logs and network traffic data should be captured and analyzed to trace the attack source and understand its impact. A clear communication plan should be in place to notify IT staff and affected users. After containment, a post-incident review should identify improvements to security procedures.
Recover	Critical network services should be restored first once the threat is fully contained, followed by non-critical services in a controlled manner. All system logs, traffic captures, and forensic data from the incident should be preserved for further analysis. The firewall configuration and network security policies should be reviewed and updated before full restoration. Recovery procedures should be documented, tested, and communicated to all relevant staff to ensure faster and more effective recovery in future incidents.

Reflections/Notes: This incident highlights the critical importance of properly configuring network security devices before deployment. A single unconfigured firewall created a vulnerability that allowed a full DoS attack to succeed. Applying the NIST CSF framework proactively — rather than reactively — would have helped identify and close this gap before the attack occurred. Going forward, regular audits of all firewall and network device configurations should be scheduled as part of standard security operations.